

$$y(t) = C_1 e^t + C_2 e^{-t} \quad \text{--- (1)}$$

$$y'(t) = C_1 e^t - C_2 e^{-t} \quad \text{--- (2)}$$

$$y''(t) = C_1 e^t + C_2 e^{-t} \quad \text{--- (3)}$$

$$\therefore C_1 e^t + C_2 e^{-t} + m(C_1 e^t - C_2 e^{-t}) + n(C_1 e^t + C_2 e^{-t}) = 0$$

$$C_1 e^t (1+m+n) + C_2 e^{-t} (1-m+n) = 0 \quad \text{--- (4)}$$

$$\therefore m+n+1=0$$

$$m+n=-1$$

As (4) holds true $\forall$

~~$C_1, C_2 \in \mathbb{R}$~~ C_1 and t